

Learning Stable State-Dependent Variable Impedance Control for Compliant Manipulation

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Abstract—State-dependent variable impedance control (VIC) has been shown as the necessary skill for compliant and dexterous robot operations. However, introducing a variable impedance law will raise the stability issue of the robot system, which introduces potential safety risks. Unfortunately, there is not yet a generally effective framework to handle this issue, i.e., existing approaches achieve stability by sacrificing the VIC performance. This article presents a learning-based VIC approach to maintain stability and control performance simultaneously. Specifically, we design a neural energy function called SNEUM, which has a globally unique minimum. Then, the SNEUM is used to encode the variable stiffness behaviors to result in stable and learnable VIC structures. Due to the special form of the SNEUM, we show that the control modes focusing on control smoothness or accuracy could be mathematically determined by tuning a hyper-parameter of the SNEUM. Various comparative experiments are conducted to show the effectiveness of the proposed approach.

Index Terms—Robotic control and learning, stability certificate learning, variable impedance control (VIC).

I. INTRODUCTION

IMPEDANCE control [1] is one of the most popular control approaches for contact-rich robotic applications. However, with robots' tendency to walk towards unstructured working environments especially in safety-sensitive human–robot collaboration tasks [2], [3], traditional impedance control is no longer capable due to its limited flexibility. Consequently, the variable impedance control (VIC), which combines impedance varying

and impedance control, has been shown as a useful approach for completing complex tasks and thus draws great attention from the robotic research society [4], [5].

The kernel work for applying VIC is the design of the variable impedance law. Online optimization approaches are always implemented to design variable impedance laws. Model predictive control (MPC) approach might be one of the most popular online optimization approaches in the robotic research society. In [6], [7], [8], the impedance model was written as a state space model, and the MPC approach was utilized to design the variable impedance law for finding an optimal accuracy-compliance trade-off. In [9], a novel nonlinear impedance model, where the impedance parameters are directly served as the model input, was formulated to allow users to consider the impedance constraints during the impedance variation process. However, when the approach considers a multistep model prediction case, the resulting MPC optimization problem is not convex, which results in a locally optimal variable impedance law. In [10], the MPC optimization problem was formulated as a robust optimization form, and based on some assumptions, the resulted variable impedance law is globally optimal and has a closed form.

Reinforcement learning (RL) is another online optimization approach, and it is sometimes more flexible and powerful due to its model-free learning abilities. In [11], a novel actor–critic algorithm was proposed to design the variable impedance laws to optimize the robot operation performances of contact tasks in uncertain environments. In [12] and [13], the policy improvement with path integrals (PI²) [14] approach was used to design optimal variable impedance laws integral with the dynamic movement primitive (DMP) [15] framework for improving the trajectory-tracking and environment-adaptive abilities. However, due to incorporating the DMP framework, the derived variable impedance laws are time-dependent, which could severely reduce their adaptability. In [16], a modified PI² approach was proposed where the DMP framework is replaced by the SEDS [17] framework, which allows the obtained variable impedance law to be fully state-dependent and hence improves the adaptability. However, all RL algorithms need to be learned via trial-and-errors, which might introduce safety issues, especially at the initial stage of the learning process.

Learning from demonstration (LfD) is another popular learning-based approach for designing variable impedance laws. The purpose is to transmit human-like impedance variations to robots. However, human impedance parameters can not be

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easily obtained via traditional sensors, and thus one of the critical challenges of LfD-based variable impedance design is to obtain the label data, i.e., the human impedance parameters. Gaussian mixture model (GMM) combined with the least square approach is popular to extract human impedance variations through the human demonstrated force, position, and velocity trajectories [18], [19]. Using the same data, a nonparameter learning approach based on a human arm dynamic model was proposed in [20] to further increase the accuracy of the human impedance estimations. In [21], the positive-definite structure of the human impedance parameters was further considered to enhance the learning performance.

Although using variable impedance laws can significantly improve the flexibility of the impedance control, it will raise the stability issues of the robot system [22], which can not be neglected for safety concerns. However, compared with the works focusing on the variable impedance law design, research on the VIC's stability guarantee is much rarer. One of the most popular approaches for stability guarantee of the VIC is tank-based approaches [22], [23]. Tank-based approaches establish a virtual tank and ensure the passivity of the robot-tank integral system by online reshaping the variable impedance law. However, the major disadvantage of this approach is that the reshaping process is not predictable and controllable, which might severely reduce the VIC performance. Moreover, the tank-based approaches are unsuitable for stability analyses in the contact-free cases, and they are sensitive to the parameter settings, such as the constant stiffness which should be used when the tank is exhausted, tank lower and upper bounds, etc. In [24], a novel Lyapunov candidate function (LCF) was designed to derive stability constraints for the variable impedance laws. However, the derived constraints will limit the flexibility of the variable impedance laws. What is more limited is that this approach is only suitable for time-dependent variable impedance laws since the derived constraints contain time derivatives of the stiffness and damping matrices. In [25], a passivity-preservation control approach was presented to ensure compliant robots' stability based on Lyapunov theorems. Similar to [24], the LCF-derived constraints will limit the flexibility of the variable impedance law. Furthermore, the constraints contain robot dynamic parameters, which are always unknown to designers and thus might further limit their applications.

Actually, both tank-based and LCF-based approaches are energy-function-based approaches. The shortcomings of these stability-guarantee approaches can be viewed as the results of choosing the inappropriate energy functions. Jin et al. [26] presented an approach to learn a neural energy function with a unique minimum, called NEUM, which has been shown as a flexible stability certificate for LfD approaches. In this article, we modify the NEUM to make it a proper VIC stability certificate, and present a stable variable impedance law learning approach accordingly. Specific contributions of this article are listed as follows.

- 1) We derive sufficient conditions to make an energy function a proper stability certificate for the state-dependent VIC. These conditions can simultaneously ensure the globally asymptotic stability and passivity of the robot

system under the contact-free and contact-rich cases, respectively.

- 2) We design a novel neural energy function called symmetric NEUM (SNEUM) based on our previous work [26]. The SNEUM has a globally unique minimum and satisfies the derived stability certificate conditions. Moreover, it is symmetric for the input sign, which avoids the influences of the tracking error sign to the variable impedance behaviors.
- 3) Two learnable VIC structures, i.e., coupled and decoupled VIC structures, are derived by encoding the desired variable stiffness behaviors via SNEUM. We show that if SNEUM parameters satisfy the proposed constraints, the closed-loop robot system will always be stable.
- 4) Thorough analyses are conducted for the SNEUM-encoded stiffness variations. We show that the variable stiffness tendency reacting from the tracking error could be mathematically determined by tuning a hyperparameter of the SNEUM, which could be a favorable means to introduce designer preferences into the learning procedure.

The proposed VIC learning approach is validated via contact-free and comparative contact-rich experiments.

II. BACKGROUND ON IMPEDANCE CONTROL

Consider the following Cartesian robot dynamic model:

$$M(x)\ddot{x} + C(x, \dot{x})\dot{x} + f_g(x) = f_c + f_e \quad (1)$$

where $M(x)$, $C(x, \dot{x}) \in R^{d_x \times d_x}$ are the inertial and Coriolis/centrifugal matrices, respectively. Vectors x , $f_g(x)$, f_c , and $f_e \in R^{d_x}$ are the robot Cartesian pose, gravity, control force, and environment-contact force, respectively. The goal of the impedance control is to reshape the dynamic model (1) into the following impedance model:

$$B_d\ddot{e} + D_d\dot{e} + K_de = f_e \quad (2)$$

where error $e = x - x_r$, and x_r is the reference robot pose. Matrices B_d , D_d , and $K_d \in R^{d_x \times d_x}$ are the desired inertial, damping, and stiffness matrices, respectively. This transformation can be achieved by designing the control force f_c as follows:

$$f_c = -(MB_d^{-1}D_d - C)\dot{e} - MB_d^{-1}K_de + (MB_d^{-1} - I)f_e + M\ddot{x}_r + C\dot{x}_r + f_g \quad (3)$$

where variables for $M(x)$, $C(x, \dot{x})$, and $f_g(x)$ are omitted for the clarity, e.g., $M \triangleq M(x)$.

Observing (3) we can find that the impedance controller is constructed by a feedback term and a compensation term, which are listed on different lines. By forcing slight constraints

$$\begin{cases} B_d = M \\ D_d = C + \bar{D}_d \end{cases} \quad (4)$$

where $\bar{D}_d$ is the designable part of the desired damping matrix, the controller could be simplified as

$$f_c = -\bar{D}_d \dot{e} - K_d e + M \ddot{x}_r + C \dot{x}_r + f_g \quad (5)$$

which is a combination of a simple proportional-derivative (PD) controller and a feed-forward compensation term. Another practically useful point is that we could implement the impedance controller without the force/torque sensor, since f_e is eliminated in the controller (5). Substituting (4) into (2), the impedance model will be

$$M \ddot{e} + (C + \bar{D}_d) \dot{e} + K_d e = f_e. \quad (6)$$

The impedance model (6) has been widely used in applications like surgical robots [22] and quadrupedal robots [10], where force/torque sensors are difficult to be installed. Note that when the velocity $\dot{x}$ is slow, the effect of C in (6) can be neglected.

It can be shown that the impedance systems (2) and (6) are passive with respect to the input–output pair $(\dot{x}, f_e)$. Since the proof for the former is naive, we only briefly introduce the passivity proof of the system (6). Defining an energy function

$$V = \frac{1}{2} \dot{e}^T M \dot{e} + \frac{1}{2} e^T K_d e \quad (7)$$

we can compute the derivative of V as follows:

$$\dot{V} = \dot{e}^T M \ddot{e} + \frac{1}{2} \dot{e}^T \dot{M} \dot{e} + \dot{e}^T K_d e. \quad (8)$$

Substituting (6) into (8) and using the property that $\dot{M} - 2C$ is skew symmetric [27], we have

$$\dot{V} = \dot{e}^T f_e - \dot{e}^T \bar{D}_d \dot{e}. \quad (9)$$

If $\bar{D}_d$ is at least semi-positive definite, we will have the following passivity condition [22]:

$$V(t) - V(0) \leq \int_0^t \dot{e}^T(t) f_e(t) dt. \quad (10)$$

Moreover, according to (9) and LaSalle's invariance principle [28], we can further state that the equilibrium point of system (6) is globally asymptotically stable when the environment-contact force $f_e = 0$.

III. PROBLEM STATEMENT

In this article, we consider a state-dependent VIC problem. Specifically, K_d and $\bar{D}_d$ in (6) will be state-dependent, i.e., they are actually two matrix-functions with forms $K_d(e)$ and $\bar{D}_d(e, \dot{e})$. Compared with the time-dependent VIC with forms $K_d(t)$ and $\bar{D}_d(t)$, the state-dependent VIC is obviously more adaptable since the impedance parameters are allowed to be online modified according to the states $(e, \dot{e})$. In the state-dependent case, the time-derivative of the energy function (7) can be computed as

$$\begin{aligned} \dot{V} &= \dot{e}^T M \ddot{e} + \frac{1}{2} \dot{e}^T \dot{M} \dot{e} + \dot{e}^T K_d(e) e + \frac{1}{2} e^T \dot{K}_d(e) e \\ &= \dot{e}^T f_e - \dot{e}^T \bar{D}_d(e, \dot{e}) \dot{e} + \frac{1}{2} e^T \dot{K}_d(e) e. \end{aligned} \quad (11)$$

For a general $K_d(e)$, the sign of term $\frac{1}{2} e^T \dot{K}_d(e) e$ is indefinite and uncontrollable, and thus we can not ensure the passivity or stability of the variable impedance system. Various approaches have been proposed to solve the stability issue, such as LCF-based approach [24] and various tank-based approaches [22], [23]. However, the LCF-based approach is theoretically designed for the time-dependent VIC. Specifically, Kronander and Billard [24] proposed constraints on the B_d , D_d , K_d , $\dot{K}_d$, and $\dot{D}_d$ to ensure the stability. Note that $\dot{K}_d$ and $\dot{D}_d$ can only be accurately obtained if K_d and D_d are time-dependent matrix functions. Although work [10] approximated $\dot{K}_d$ using the time-differential technique to utilize the approach for state-dependent VIC, the stability actually can only be approximately ensured. Moreover, the proposed constraints will limit the flexibility of the variable impedance law.

Tank-based approaches ensure the stability by designing a virtual tank. Informally speaking, when $\dot{V}$ in (11) is negative, the virtual tank will store the dissipated energy of the robot system. When $\dot{V} > 0$, the energy in the virtual tank will be extracted to support the impedance variations until the tank is exhausted, and then a designed constant stiffness will be used to reshape the desired stiffness. However, we can never predict when the tank will be exhausted, which means that the reshaping behaviors can not be predicted. Thus, a bad tank design will severely reduce the VIC performance. Moreover, the frequently reshaping behaviors will cause robot oscillations, which might raise the safety issues.

As mentioned in Section I, both LCF-based and tank-based approaches are energy-function-based approaches. The disadvantages of these existing approaches are caused by selecting the inappropriate energy functions. For example, if the energy functions V is properly designed, i.e., $\dot{V}$ is always negative under the desired variable stiffness $K_d(e)$ and $\bar{D}_d(e, \dot{e})$, the constraints in [24] or the tanks in [22] and [23] will be extra and redundant. In this article, we try to design such a proper energy function. Let us observe the form (11), the unstable or nonpassive hazard is mainly caused by the variable stiffness. Thus, we try to solve the mentioned issue by encoding the variable stiffness into an energy function. Mathematically, we are committed to design an energy function $V_2(e)$ such that the total energy

$$V = \frac{1}{2} \dot{e}^T M \dot{e} + V_2(e) \quad (12)$$

will always be decreasing along the variable impedance system. We name $V_2(e)$ as the stiffness-encoding energy function. Actually, even for the constant stiffness case, the second term is also the key to compensate the stiffness-related energy variation, which can be found in the derivation from (8) to (9). Moreover, in the constant stiffness case, the stiffness-encoding energy function will be $V_2 = e^T K_d e$, see (7).

The remaining problem is how to design the stiffness-encoding energy function V_2 such that the desired variable impedance system is passive ($f_e \neq 0$) and globally asymptotically stable ($f_e = 0$). In this article, we assume that a desired variable-impedance set $\Psi = \{e_t, \dot{e}_t, K_{d,t}, \bar{D}_{d,t}\}$ has been provided, where $t \in [1, \dots, T]$ is the index of a data-pair, and T is the size of the set Ψ . The set Ψ could be generated from an original desired variable impedance law, or be collected from human

demonstrations, or even be directly set by an experienced human tutor. We will learn V_2 based on the set Ψ , and then generate the desired stiffness $K_d(e)$ and damping $\bar{D}_d(e, \dot{e})$ accordingly. In the next section, we will analyze the condition for V_2 such that the total energy function (12) could be an effective passivity and stability certificate for the variable impedance system.

IV. PASSIVITY/STABILITY ANALYSIS WITH THE STIFFNESS-ENCODING ENERGY FUNCTION

In this section, we analyze the passivity (stability) of the state-dependent variable impedance system at the case $f_e \neq 0$ ($f_e = 0$). To keep the self-contained structure, we give the state-dependent variable impedance system and the energy function as follows:

$$\begin{cases} (a). & M\ddot{e} + (C + \bar{D}_d(e, \dot{e}))\dot{e} + K_d(e)e = f_e \\ (b). & V = \frac{1}{2}\dot{e}^T M \dot{e} + V_2(e) \end{cases} \quad (13)$$

where $V_2(e)$ is the stiffness-encoding energy function. We can compute $\dot{V}$ as follows:

$$\begin{aligned} \dot{V} &= \dot{e}^T M \ddot{e} + \frac{1}{2}\dot{e}^T \dot{M} \dot{e} + \dot{e}^T \frac{\partial V_2(e)}{\partial e} \\ &= \dot{e}^T f_e - \dot{e}^T \bar{D}_d(\dot{e}, e)\dot{e} + \dot{e}^T \left(\frac{\partial V_2(e)}{\partial e} - K_d(e)e \right) \end{aligned} \quad (14)$$

where the second row can be obtained by using (13.a) and the property that $\dot{M} - 2C$ is skew symmetric. Arrive here we can mathematically give the stiffness encoding form as

$$\frac{\partial V_2(e)}{\partial e} = K_d(e)e. \quad (15)$$

We first consider the passivity of the variable impedance system (13.a) at the case $f_e \neq 0$. It is obvious that the system will be passive if $\bar{D}_d(\dot{e}, e)$ and $V_2(e)$ both be at least semi-positive definite.

We then consider the globally asymptotic stability of the variable impedance system (13.a) at the case $f_e = 0$. Substituting $f_e = 0$ and (15) into (13.a), we have

$$M\ddot{e} + (C + \bar{D}_d(\dot{e}, e))\dot{e} + \frac{\partial V_2(e)}{\partial e} = 0. \quad (16)$$

Assuming $\bar{D}_d(\dot{e}, e)$ is positive definite, the semi-negative definite function V can take value 0 only at the case $\dot{e} = 0$. Substituting $\dot{e} \equiv 0$ into (16), we have

$$\frac{\partial V_2(e)}{\partial e} = 0. \quad (17)$$

Thus, according to the LaSalle's invariance principle [28], we can state that the equilibrium of the variable impedance system (16), or equivalently (13.a), is globally asymptotically stable if $V_2(e)$ is positive definite, radially unbounded and has the unique-minimum property

$$\begin{cases} \frac{\partial V_2(e)}{\partial e} = 0, e = 0 \\ \frac{\partial V_2(e)}{\partial e} \neq 0, e \neq 0. \end{cases} \quad (18)$$

We conclude the main results of this section as the following theorem.

Theorem 1: The state-dependent variable impedance system (13.a) is passive with respect to $(\dot{e}, f_e)$ if $\bar{D}_d(\dot{e}, e)$ is at least semi-positive definite, and $K_d(e)e = \frac{\partial V_2(e)}{\partial e}$ where $V_2(e)$ is differentiable and at least semi-positive definite. Moreover, if $\bar{D}_d(\dot{e}, e)$ and $V_2(e)$ are further positive definite, and $V_2(e)$ is also radially unbounded and satisfies the unique-minimum condition (18), the equilibrium point of the system (13.a) is globally asymptotically stable (GAS) in the case $f_e = 0$.

V. DESIGNING THE STIFFNESS-ENCODING ENERGY FUNCTION

According to the Theorem 1, the best choice of the stiffness-encoding energy function $V_2(e)$ is a positive-definite, differentiable, radially unbounded function with the unique-minimum property (18). In this case, with a positive-definite $\bar{D}_d(\dot{e}, e)$, the state-dependent variable impedance system (13.a) will be passive and globally asymptotically stable for $f_e \neq 0$ and $f_e = 0$, respectively. In this section, we aim to construct such a type of functions.

Lemma 1: The energy function $\Gamma(e)$ with the form

$$\begin{cases} \Gamma(e) = J_1(e) - J_1(0) + J_2(e) \\ J_1(e) = \omega^T f(e) \\ f(e) = [f_1(e), \dots, f_k(e), \dots, f_{d_H}(e)]^T \\ f_k(e) = \varrho(a_k^T z(e) + b_k) \\ z(e) = [\|e\|_2^{1+\epsilon_1} \quad \|e\|_2^{\epsilon_1} e^T]^T \\ J_2(e) = \alpha e^T e \end{cases} \quad (19)$$

where $\omega \in R^{d_H}$, $\varrho(s) = \frac{e^s - e^{-s}}{e^s + e^{-s}}$, $a_k \in R^{d_x+1}$, $b_k \in R$, $\epsilon_1, \alpha \in R_{++}$, is positive-definite, continuously-differentiable, radially unbounded, and has the unique-minimum property

$$\begin{cases} \frac{\partial \Gamma(e)}{\partial e} = 0, e = 0 \\ \frac{\partial \Gamma(e)}{\partial e} \neq 0, e \neq 0 \end{cases} \quad (20)$$

if the following constraints hold:

$$\begin{cases} a_{k,1} > 0 \\ a_{k,1}^2 - \sum_{i=2}^{d_x+1} a_{k,i}^2 > 0, \forall k \in [1, \dots, d_H] \\ \omega_k > 0 \end{cases} \quad (21)$$

where $a_{k,i}$ is the i th component of the vector a_k , and ω_k is the k th component of the vector ω . Moreover, $\Gamma(e)$ is monotonically increasing along any rays emitted from the origin.

Proof: See Propositions 3 and 4 in our previous work [26].

The energy function $\Gamma(e)$ in (19) is named as the “NEUM,” which refers to a type of neural energy functions with a unique minimum. It has been shown in our previous work [26] that the constraints (21) will not introduce too much limitations for the model flexibility. Although we may directly use $\Gamma(e)$ as the stiffness-encoding energy function, we should note that $\Gamma(e)$ is not symmetric, e.g., $\Gamma(-e) \neq \Gamma(e)$. It will indeed be a problem since in this case, the $K_d(e)$ will be influenced by the sign of the error e , which is not reasonable. A familiar symmetric energy function is the quadratic function $e^T K_d e$ which corresponds to the constant stiffness as mentioned in Section III. To keep the symmetric property, we slightly modify the function $\Gamma(e)$ to

$\bar{\Gamma}(e)$ as

$$\bar{\Gamma}(e) = \Gamma(|e|^{1+\epsilon_2}) \quad (22)$$

where ϵ_2 is a positive scalar, $|\cdot|^{1+\epsilon_2}$ is an element-wise operation, i.e.,

$$|e|^{1+\epsilon_2} = \begin{bmatrix} |e_1|^{1+\epsilon_2} & \dots & |e_i|^{1+\epsilon_2} & \dots & |e_{d_x}|^{1+\epsilon_2} \end{bmatrix}^T. \quad (23)$$

The function $\bar{\Gamma}(e)$ is named as the SNEUM.

Denote $h(e) = |e|^{1+\epsilon_2}$, we will show that

$$\bar{\Gamma}(e) = \Gamma(h(e)) \quad (24)$$

is also positive-definite, continuously differentiable, radially unbounded, and has the unique-minimum property. To achieve the purpose, we first give the following useful result.

Proposition 1: For any e containing zero element(s), denote $\bar{e}$ as the vector constructed by nonzero element(s) of e , we have $\frac{\partial \bar{\Gamma}(e)}{\partial \bar{e}} \neq 0$ if constraints (21) hold.

Proof: We prove the result by contradiction. Consider the ray passing through the origin and e which has zero element(s), if $\frac{\partial \bar{\Gamma}(e)}{\partial \bar{e}} = 0$, the $\Gamma(e)$ will not be monotonically increasing along this ray, which violates the Lemma 1 if constraints (21) hold. This completes the proof. ■

Then, we can state the main result of this section as follows.

Proposition 2: The SNEUM (22) is positive-definite, continuously differentiable, radially unbounded, and has the unique-minimum property

$$\begin{cases} \frac{\partial \bar{\Gamma}(e)}{\partial e} = 0, e = 0 \\ \frac{\partial \bar{\Gamma}(e)}{\partial e} \neq 0, e \neq 0 \end{cases} \quad (25)$$

if the constraints (21) hold.

Proof: According to the Lemma 1 and the definition (22), it is obvious that the SNEUM $\bar{\Gamma}$ is positive-definite and radially unbounded if constraints (21) hold. We then show the continuously differentiable and the unique-minimum properties of $\bar{\Gamma}$. Using the chain rule and definition (24), we have

$$\frac{\partial \bar{\Gamma}(e)}{\partial e} = \left(\frac{\partial h(e)}{\partial e} \right)^T \frac{\partial \Gamma(h)}{\partial h} \quad (26)$$

where

$$\frac{\partial h}{\partial e} = \left(\frac{\partial h}{\partial e} \right)^T = (1 + \epsilon_2) \text{diag}\{\dots \text{sign}(e_i) |e_i|^{\epsilon_2} \dots\} \quad (27)$$

$\text{diag}\{\dots \delta_i \dots\}$ represents a diagonal matrix with elements δ_i , and

$$\text{sign}(e_i) = \begin{cases} -1, e_i \leq 0 \\ 1, e_i > 0. \end{cases} \quad (28)$$

Since $\frac{\partial \Gamma(h)}{\partial h}$ is continuous and $\epsilon_2 > 0$, we can state that $\bar{\Gamma}(e)$ is continuously differentiable. As for the unique-minimum property, it is obvious that $\frac{\partial \bar{\Gamma}(e)}{\partial e} = 0$ for $e = 0$. As for $e \neq 0$, we first consider the case that e does not contain any zero element, i.e., $e_i \neq 0$ for $\forall i \in [1, \dots, d_x]$. In this case, the matrix $\frac{\partial h}{\partial e}$ is full-rank, and we have $h(e) \neq 0$. According to the Lemma 1, we further have $\frac{\partial \Gamma(h)}{\partial h} \neq 0$, and thus $\frac{\partial \bar{\Gamma}(h)}{\partial e} \neq 0$.

We then consider the case that e contains zero element(s). Denote $\bar{e}$ as the vector constructed by the nonzero elements of e , then $\bar{h}(\bar{e}) = |\bar{e}|^{1+\epsilon_2}$ is the vector constructed by the nonzero elements of $h(e)$. According to the Proposition 1, we have $\frac{\partial \Gamma(h)}{\partial h} \neq 0$. Moreover, matrix $\frac{\partial \bar{h}(\bar{e})}{\partial \bar{e}}$, which is the nonzero part of the diagonal matrix $\frac{\partial h(e)}{\partial e}$, is obviously full-rank. Thus, we have

$$\left(\frac{\partial \bar{h}(\bar{e})}{\partial \bar{e}} \right)^T \frac{\partial \Gamma(h)}{\partial h} \neq 0 \quad (29)$$

which implies that part of $\frac{\partial \bar{\Gamma}(e)}{\partial e}$'s elements are not zero. This completes the proof. ■

The SNEUM $\bar{\Gamma}(e)$ can then be used as the stiffness-encoding energy function, i.e., $V_2(e) \triangleq \bar{\Gamma}(e)$.

Remark 1: From (27) we can find that the parameter ϵ_2 is used to ensure the existence of the gradient $\frac{\partial \Gamma(h)}{\partial h}$ at cases with $e_i = 0$, which ensures the smooth property of the SNEUM.

VI. LEARNING COUPLED STATE-DEPENDENT VIC

In this section, we show how to learn the coupled state-dependent VIC utilizing the SNEUM $\bar{\Gamma}(e)$ proposed in the last section. The word “coupled” means that $K_d(e)$ and $\bar{D}(\dot{e}, e)$ are generally not diagonal.

A. Learning Coupled Variable Impedance Behaviors

Learning a positive-definite $\bar{D}_d(\dot{e}, e)$ based on the set Ψ defined in the Section III is trivial, such as using tensor-based approach [21]. Thus, we step the details for this in the section.

The most important part of the variable impedance controller is the variable stiffness part. As mentioned in Sections IV and V, the variable stiffness $K_d(e)$ has the form

$$\begin{cases} \frac{\partial V_2(e, \Theta)}{\partial e} = K_d(e, \Theta) e \\ V_2(e, \Theta) = \bar{\Gamma}(e, \Theta) \end{cases} \quad (30)$$

where the SNEUM $\bar{\Gamma}(e, \Theta)$ is defined as (22). Note that the $\Theta \triangleq \{a_k, b_k, \omega_k\}$ is also regarded as the variable for functions in (30) since it is served as the learnable parameter in this section. Thus, in order to get $K_d(e, \Theta)$, we first need to learn the $V_2(e, \Theta)$, or equivalently the $\bar{\Gamma}(e, \Theta)$. The objective function of the learning problem is formulated as follows:

$$J(\Theta) = \frac{1}{T} \sum_{t=1}^T \left\| \frac{\partial V_2(e_t, \Theta)}{\partial e_t} - K_{d,t} e_t \right\|^2 \quad (31)$$

where T is the size of the set Ψ , e_t and $K_{d,t}$ are the set samples, see definitions in Section III. Moreover, we should also consider the constraints (21) to ensure that the $V_2(e)$ is an effective stiffness-encoding energy function. Thus, the learning problem is formulated as follows:

$$\min_{\Theta} J(\Theta) \quad (32)$$

subject to

$$\begin{cases} (a). & a_{k,1} > 0 \\ (b). & a_{k,1}^2 - \sum_{i=2}^{d_H+1} a_{k,i}^2 > 0, \forall k \in [1, \dots, d_H] \\ (c). & \omega_k > 0 \end{cases} \quad (33)$$

Once we obtain the optimal Θ^* by solving the above learning problem, we can compute the $V_2(e, \Theta^*)$ and $K_d(e, \Theta^*)$ using (30). As a conclusion, the decoupled state-dependent variable impedance controller is given as follows:

$$\begin{aligned} f_c &= -\bar{D}_d(e, \dot{e})\dot{e} - K_d(e, \Theta^*)e + M\ddot{x}_r + C\dot{x}_r + f_g \\ &= -\bar{D}_d(e, \dot{e})\dot{e} - \frac{\partial V_2(e, \Theta^*)}{\partial e} + M\ddot{x}_r + C\dot{x}_r + f_g. \end{aligned} \quad (34)$$

Actually, in practically implementation, we need not to compute the closed-form of $K_d(e, \Theta)$, which can be seen in the second row of (34). However, it is worthwhile to derive $K_d(e, \Theta)$ in order to better interpret the coupled state-dependent variable impedance controller.

B. Analysis for the Coupled Variable Stiffness Behaviors

In this section, we derive the closed-form of $K_d(e)$ to analyze the variable stiffness behaviors. Based on (26) and (27), what remains is to compute $\frac{\partial \Gamma(h)}{\partial h}$. According to the chain rule, we have

$$\begin{cases} \frac{\partial \Gamma(h)}{\partial h} = \frac{\partial J_1(h)}{\partial h} + 2\alpha h \\ \frac{\partial J_1(h)}{\partial h} = \left(\frac{\partial z(h)}{\partial h} \right)^T \left(\frac{\partial f(z)}{\partial z} \right)^T \omega \end{cases} \quad (35)$$

where we reclaim that h is the modified error defined as $h(e) = [e_1^{1+\epsilon_2} \dots |e_i|^{1+\epsilon_2} \dots |e_{d_H}|^{1+\epsilon_2}]^T$, and

$$\begin{cases} \left(\frac{\partial f(z)}{\partial z} \right)^T = [\dots (1 - \varrho_k^2) a_k \dots], k \in [1, \dots, d_H] \\ \left(\frac{\partial z(h)}{\partial h} \right)^T = \left[(1 + \epsilon_1) \|h\|_2^{\epsilon_1} \frac{h}{\|h\|_2} \quad \epsilon_1 \|h\|_2^{\epsilon_1} \frac{h h^T}{\|h\|_2^2} + \|h\|_2^{\epsilon_1} I \right] \\ \varrho_k \triangleq \varrho(a_k^T z(h) + b_k) \end{cases} \quad (36)$$

We rewrite $\left(\frac{\partial z(h)}{\partial h} \right)^T$ in (36) as follows:

$$\left(\frac{\partial z(h)}{\partial h} \right)^T = h \|h\|_2^{\epsilon_1-1} \left[(1 + \epsilon_1) \quad \epsilon_1 \frac{h^T}{\|h\|_2} \right] + \|h\|_2^{\epsilon_1} \begin{bmatrix} 0 & I \end{bmatrix}. \quad (37)$$

Denote the first and second terms in (37) as $\left(\frac{\partial z(h)}{\partial h} \right)_1^T$ and $\left(\frac{\partial z(h)}{\partial h} \right)_2^T$, respectively, we have

$$\begin{aligned} & \left(\frac{\partial z(h)}{\partial h} \right)_1^T \left(\frac{\partial f(z)}{\partial z} \right)^T \omega \\ &= \sum_{k=1}^{d_H} \omega_k (1 - \varrho_k^2) \left(a_{k,1} + \epsilon_1 \left(a_{k,1} + \frac{h^T a_{k,2:}}{\|h\|_2} \right) \right) \|h\|_2^{\epsilon_1-1} h \end{aligned} \quad (38)$$

where $a_{k,2:}$ is the vector constructed by the second to last components of a_k , and

$$\left(\frac{\partial z(h)}{\partial h} \right)_2^T \left(\frac{\partial f(z)}{\partial z} \right)^T \omega = \sum_{k=1}^{d_H} \omega_k (1 - \varrho_k^2) \frac{a_{k,2:} h^T}{\|h\|_2} \|h\|_2^{\epsilon_1-1} h. \quad (39)$$

Substituting (38) and (39) into (35), we have

$$\frac{\Gamma(h)}{\partial h} = (G(h) + g(h)I + 2\alpha) h \quad (40)$$

where I is the identity matrix, and

$$\begin{cases} g(h) = \epsilon_1 \|h\|_2^{\epsilon_1-1} \sum_{k=1}^{d_H} \omega_k (1 - \varrho_k^2) \left(a_{k,1} + \frac{h^T a_{k,2:}}{\|h\|_2} \right) \\ G(h) = \|h\|_2^{\epsilon_1-1} \sum_{k=1}^{d_H} \omega_k (1 - \varrho_k^2) \left(a_{k,1} I + \frac{a_{k,2:} h^T}{\|h\|_2} \right) \end{cases} \quad (41)$$

Let us observe the gains $g(h)$ and $G(h)$ in (41). If constraints (33.a) and (33.b) hold, we have

$$a_{k,1} + \frac{h^T a_{k,2:}}{\|h\|_2} > 0, \forall h. \quad (42)$$

Combine with the constraint (33.c), we claim that $g(h) > 0$ for any h . As for the matrix gain $G(h)$, the definite property may not be easily determined. However, we will show that it is always positive in the decoupled case, which will be introduced in the next section.

Moreover, due to the existence of the term $\|h\|_2^{\epsilon_1-1}$, the relationships between gain values and the modified error norm $\|h\|_2$ in area around $h = 0$ will be determined by the parameter ϵ_1 . Specifically, when $0 < \epsilon_1 < 1$, gain values will approach infinity when $h \rightarrow 0$, and be negatively related with the $\|h\|_2$ around $h = 0$. On the contrary, when $\epsilon_1 > 1$, gain values equal 0 when $h = 0$, and be positively related with the $\|h\|_2$ around $h = 0$. These two modes have their own advantages. Specifically, if $0 < \epsilon_1 < 1$, the tendency that gain values approach to infinity when $h \rightarrow 0$ implies that the controller will generate persistent force to eliminate small tracking errors. Otherwise, if $\epsilon_1 > 1$, the control signal will be more smooth around $h = 0$. We will validate these properties in experiments in detail.

We then step forward to the $K_d(e)$ derivation. Substituting (41) into (26), we can obtain

$$\frac{\partial \bar{\Gamma}(e)}{\partial e} = (1 + \epsilon_2) \Lambda (G(h) + (g(h) + 2\alpha) I) h \quad (43)$$

where

$$\Lambda = \text{diag}\{\dots \text{sign}(e_i) |e_i|^{\epsilon_2} \dots\}. \quad (44)$$

Since h can be expanded as $h = \Lambda e$, we have

$$\frac{\partial \bar{\Gamma}(e)}{\partial e} = (1 + \epsilon_2) (\Lambda^2 (g(h) + 2\alpha) + \Lambda G(h) \Lambda) e. \quad (45)$$

According to (30) we can obtain

$$K_d(e) = (1 + \epsilon_2) (\Lambda^2 (g(h) + 2\alpha) + \Lambda G(h) \Lambda). \quad (46)$$

Remark 2: It should be noted that the functions h and Λ in (46) are all e -dependent, and thus K_d is also e -dependent.

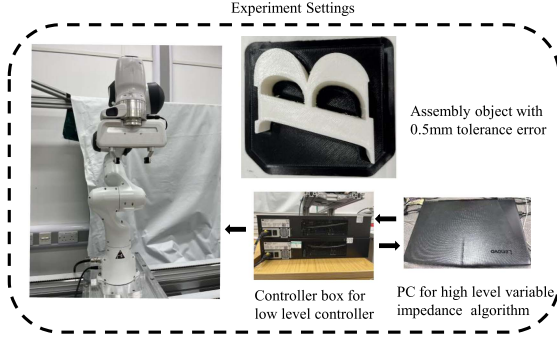


Fig. 1. Robot assembly task.

The constant $\epsilon_2 \in R_{++}$ is only used to ensure the differential ability of $\bar{\Gamma}(e)$ at cases $e_i = 0$. Thus, it could be taken as an extremely small positive value, which implies that $\Lambda^2 \approx I$ and $\Lambda^2 g(h) \approx g(h)I$. As for the term $\Lambda G(g)\Lambda$, we can state that the amplitudes of its elements are approximately same as these of $G(h)$ if ϵ_2 is extremely small. Thus, terms $\Lambda^2 g(h)$ and $\Lambda G(h)\Lambda$ share the same properties of $g(h)$ and $G(h)$.

Arrive here we have given a coupled state-dependent variable impedance controller, under which the stability and passivity could be simultaneously ensured. We conclude this result as the following theorem.

Theorem 2: The variable impedance system (13.a) is, respectively, GAS and passive with respect to $(\dot{e}, f_e)$ at the cases $f_e = 0$ and $f_e \neq 0$, if $\bar{D}_d(e, \dot{e})$ is any positive-definite matrix function, constraints (21) hold, and $K_d(e)$ has the form (46) where $h(e) = [|e_1|^{1+\epsilon_2} \quad \dots \quad |e_i|^{1+\epsilon_2} \quad \dots \quad |e_{d_x}|^{1+\epsilon_2}]^T$, ϵ_2 and α are any positive constants, $\Lambda(e)$ is given by (44), $g(h(e))$ and $G(h(e))$ are defined in (41).

VII. LEARNING DECOUPLED STATE-DEPENDENT VIC

In this section, we aim to learn a decoupled state-dependent variable impedance controller, i.e., $K_d(e)$ and $\bar{D}_d(\dot{e}, e)$ follow the forms:

$$\begin{cases} K_d(e) = \text{diag}\{\dots k_{d,i}(e_i) \dots\} \\ D_d(\dot{e}, e) = \text{diag}\{\dots d_{d,i}(\dot{e}_i, e_i) \dots\} \end{cases} \quad (47)$$

Actually, the decoupled forms (47) are more consistent with the human control style. It is also more convenient to provide diagonal K_d and $\bar{D}_d$ in set Ψ for a human tutor.

A. Learning Decoupled Variable Impedance Behaviors

The learning details for $D_d(\dot{e}, e)$ are also omitted as in last section since it is trivial, and we mainly introduce the learning approach for $K_d(e)$, or equivalently for each $k_{d,i}(e_i)$. Different from the coupled case, the stiffness-encoding energy function for the decoupled case is designed as

$$V_2(e) = \sum_{i=1}^{d_x} \bar{\Gamma}_i(e_i) \quad (48)$$

where function $\bar{\Gamma}_i(\cdot)$ is the SNEUM introduced in the Section V. According to (48) and the stiffness-encoding form (15), we have

$$k_{d,i}(e_i, \Theta_i) = \frac{\partial \bar{\Gamma}_i(e_i, \Theta_i)}{\partial e_i} \quad (49)$$

where $\Theta_i \triangleq \{a_{i,k}, b_{i,k}, \omega_{i,k}\}$ is the parameter set expected to be learned. Note that we regard Θ_i as the variable for functions in (49), since it will be learned in this section. The learning problem for each $k_{d,i}$ is formulated as

$$\min_{\Theta_i} J(\Theta_i) = \frac{1}{T} \sum_{i=t}^T \left\| \frac{\partial V_2(e_{t,i}, \Theta_i)}{\partial e_{t,i}} - k_{d,t,i} e_{t,i} \right\|^2 \quad (50)$$

subject to

$$\begin{cases} (a). & a_{i,k,1} > 0 \\ (b). & a_{i,k,1}^2 - a_{i,k,2}^2 > 0, \forall k \in [1, \dots, d_H] \\ (c). & \omega_{i,k} > 0 \end{cases} \quad (51)$$

where $e_{t,i}$ and $k_{d,t,i}$ can be obtained from the t th sample of the set Ψ . Once we get each optimal Θ_i^* , we can finally obtain the decoupled variable impedance controller as follows:

$$f_c = -\bar{D}_d(e, \dot{e})\dot{e} - \sum_{i=1}^{d_x} \frac{\partial \bar{\Gamma}_i(e_i, \Theta_i^*)}{\partial e_i} + M\ddot{x}_r + C\dot{x}_r + f_g. \quad (52)$$

Similar with coupled case, we need not to compute each $k_{d,i}(e_i)$ when we implement the decoupled controller (52). However, we claim that it is worthwhile to derive it for better understanding the variable stiffness behaviors.

B. Analysis for the Decoupled Variable Stiffness Behaviors

In this section, we derive the closed-form of $k_{d,i}(e_i)$ to intuitively interpret the variable stiffness behaviors. The result can be directly obtained based on the derivations of Section VI-B. Since we focus the i th dimension, we need to add a subscript “i” for symbols in Section VI-B. For example, (40) will be

$$\frac{\Gamma_i(h_i)}{\partial h_i} = (G_i(h_i) + g_i(h_i) + 2\alpha_i) h_i \quad (53)$$

where

$$\begin{cases} g_i(h_i) = \epsilon_{1,i} h_i^{\epsilon_{1,i}-1} \sum_{k=1}^{d_H} \omega_{i,k} (1 - \varrho_{i,k}^2) (a_{i,k,1} + a_{i,k,2}) \\ G_i(h_i) = h_i^{\epsilon_{1,i}-1} \sum_{k=1}^{d_H} \omega_{i,k} (1 - \varrho_{i,k}^2) (a_{i,k,1} + a_{i,k,2}) \end{cases} \quad (54)$$

Note that for the decoupled case, h_i , $g_i(h_i)$, and $G_i(h_i)$ are all scalars, and we will have $g_i(h_i) > 0$ and $G_i(h_i) > 0$ due to the constraints (51). For simplicity, (53) can be rewritten as

$$\frac{\Gamma_i(h_i)}{\partial h_i} = (\bar{g}_i(h_i) + 2\alpha_i) h_i \quad (55)$$

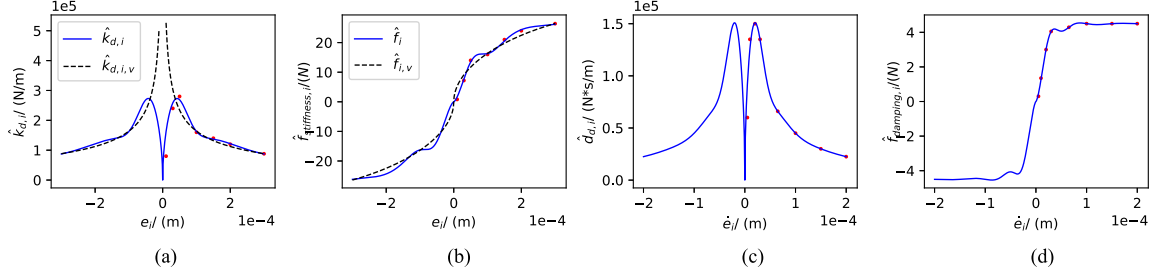


Fig. 2. Learning results of the variable impedance behaviors. (a) Stiffness encoding results under two different ϵ_1 settings. (b) Stiffness-related force encoding results under two different ϵ_1 settings. (c) Damping encoding results under the case $\epsilon_1 = 1.5$. (d) Damping-related force encoding result under the case $\epsilon_1 = 1.5$.

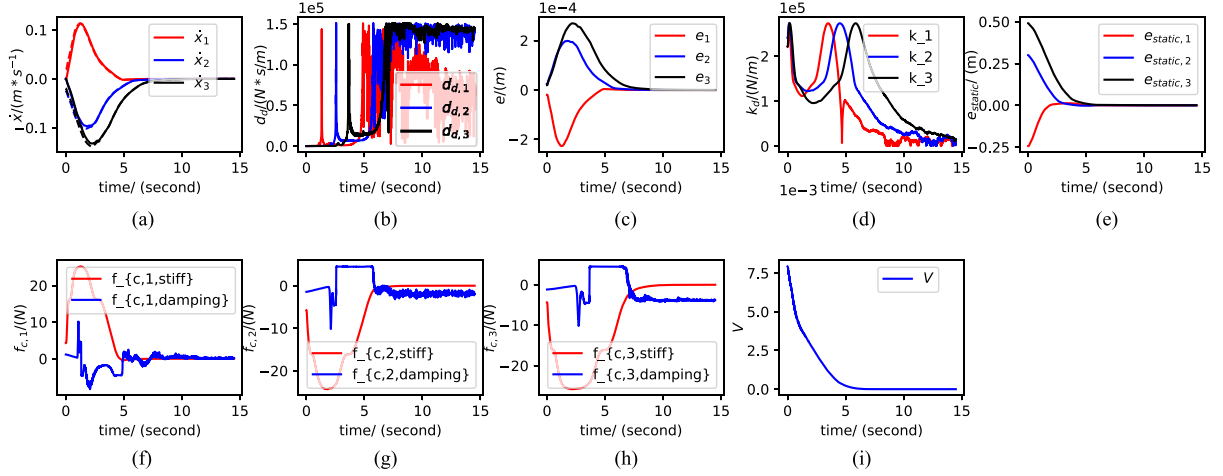


Fig. 3. Results of the robot assembly experiment using the second-order planner (59) with $\kappa = [40.0 \ 40.0 \ 20.0]^T$, and the proposed VIC with setting $\epsilon_1 = 1.5$. (a) Velocity tracking result. (b) Damping variation. (c) Position tracking error. (d) Stiffness variation. (e) Distance to the goal position. (f) Control force in the x dimension. (g) Control force in the y dimension. (h) Control force in the z dimension. (i) Energy variation.

where

$$\bar{g}_i(h_i) = (1 + \epsilon_{1,i})h_i^{\epsilon_{1,i}-1} \sum_{k=1}^{d_H} \omega_{i,k} (1 - \varrho_{i,k}^2) (a_{i,k,1} + a_{i,k,2}). \quad (56)$$

The Λ_i corresponding to (44) will also be a scalar as

$$\lambda_i \triangleq \Lambda_i = \text{sign}(e_i) |e_i|^{\epsilon_{2,i}}. \quad (57)$$

Thus, according to (46), we can finally obtain

$$k_{d,i}(e_i) = (1 + \epsilon_{2,i})\lambda_i^2 (\bar{g}_i(h_i) + 2\alpha_i) \quad (58)$$

where we restate that $h_i = |e_i|^{1+\epsilon_2}$ is a modified error for the i th dimension. Compared with the coupled case, the decoupled case is easier to be interpreted since the gain is a scalar. Note that $\bar{g}_i(h_i)$ inherits all of the properties of $g(h)$ in (41), i.e., be positive definite and has two relationship modes with h_i dependent on the value of $\epsilon_{1,i}$, see analysis in Section VI-B. Since we have $(1 + \epsilon_{2,i})\lambda_i^2 > 0$, we state that $k_{d,i}(e_i)$ has the same properties of $\bar{g}_i(h_i)$. Similar with the coupled case in last Section, we conclude the decoupled VIC controller as the following theorem.

Theorem 3: The variable impedance system (13.a) is, respectively, GAS and passive with respect to $(\dot{e}, f_e)$ at the cases $f_e = 0$

and $f_e \neq 0$, if $\bar{D}_d(e, \dot{e})$ is given as (47) with positive-definite $d_{d,i}(\dot{e}_i, e_i)$ s, constraints (21) hold, and $K_d(e)$ has the form (47) with $k_{d,i}(e_i)$ defined in (58), where $h_i(e_i) = |e_i|^{1+\epsilon_2}$, $\epsilon_{2,i}$ and α_i are any positive constants, λ_i is given by (57), and $\bar{g}_i(h_i(e_i))$ is defined in (56).

VIII. EXPERIMENTAL VALIDATIONS

In this section, the contact-free robot assembly and contact-rich robot polish tasks are conducted to show the effectiveness of the proposed approach.

A. Robot Assembly Experiment

In this section, the proposed approach is validated via the robot assembly experiment. We will thoroughly analyze the properties of the proposed state-dependent VIC approach in this experiment, and leave the comparisons with other stable VIC approaches in the robot polish experiment. The robot assembly experiment is shown in the Fig. 1, where experiment devices contain a Franka robot arm, a controller box, a PC with Core i7-7700HQ 2.8GHz CPU, and an assembly object with 0.5 mm tolerance error. In this experiment, the robot orientation is controlled via a constant impedance controller, and we adapt the proposed approach on the robot position control. The following

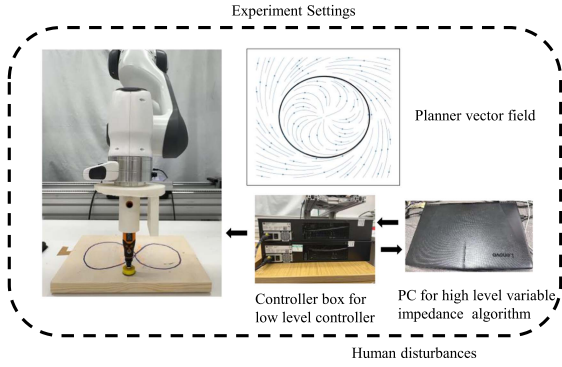


Fig. 4. Robot polish task.

decoupled second-order dynamic system is used as the robot reference position trajectory planner

$$\ddot{x}_{r,i} = -2\sqrt{\kappa_i}\dot{x}_i - \kappa_i e_{\text{static},i} \quad (59)$$

where $e_{\text{static},i} = x_i - x_{\text{goal},i}$ is the distance between the robot's current position and the goal position at the i th dimension. Note the difference between the $e_{\text{static},i}$ and e_i , where the latter is defined as $e_i = x_i - x_{r,i}$. The parameter $\kappa = [\kappa_1, \dots, \kappa_{d_x}]^T$ is used to determine the convergence rate of the planner, and it is set as $\kappa = [40.0 \ 40.0 \ 20.0]^T$ in the experiment.

Accordingly, the decoupled state-dependent variable impedance controller introduced in Section VII is used for the robot control, and the SNEUM, or the variable stiffness law, for each dimension is the same. The parameter-settings for the SNEUM are given as follows after trial-and-errors.

$$\begin{cases} d_H = 10, & \epsilon_1 = 1.5, & \alpha = 10^{-3} \\ \epsilon_2 = 10^{-10} \end{cases} \quad (60)$$

The damping terms $d_{d,i}$ s are also encoded via SNEUMs with the same parameter-settings as (60). Given the desired variable impedance set $\Psi = \{e_t, \dot{e}_t, K_{d,t}, \overline{D}_{d,t}\}$, the learning results of the variable impedance law are shown in Fig. 2.

The stiffness encoding result is shown in Fig. 2(a), where the red points represent the samples in the set Ψ , the solid and dashed lines are, respectively, the learning results under the cases $\epsilon_1 = 1.5$ and $\epsilon_1 = 0.5$. In Section VII-B, we analyze the influence of the parameter ϵ_1 to the stiffness encoding result, that is, in area near the origin, the stiffness will be infinite (zero) and negatively (positively) related the tracking error if $0 < \epsilon_1 < 1.0$ ($\epsilon > 1.0$). The learning result here is exactly what we expect. Moreover, we can also find that, although the set Ψ is more consistent with the setting $\epsilon > 1.0$, a violated choice of ϵ_1 will also useful for encoding the variable stiffness in area away from the origin.

Fig. 2(b) shows the stiffness-related force encoding result, where the stiffness-related force in the i th dimension is defined as $f_{\text{stiffness}} = k_{d,i}e_i$. The solid and dashed lines, respectively, represent the force encoding results under the cases $\epsilon_1 = 1.5$ and $\epsilon_1 = 0.5$. We may find that the force can be both well encoded under these two different cases. However, the most obvious difference between these two cases lies on the force variation near the origin. The slope of the solid line approaches zero near the origin, while the slope of the dashed line approaches infinity when e_i approaches 0. This difference is consistent with

Fig. 2(a) since the slope of the stiffness-related force is exactly the stiffness.

We state that these two different settings of ϵ_1 actually result into two different control modes. The setting $0 < \epsilon_1 < 1.0$ allows a persistent control force even though the error is very small, which can improve the tracking accuracy. Oppositely, the controller corresponding to the setting $\epsilon_1 > 1.0$ will be more smooth around the zero-error area. Moreover, if the controller is expected to be totally data-driven, we can set the $\epsilon_1 = 1.0$. Fig. 2(c) and (d) shows the similar learning contents as these in Fig. 2(a) and (b), except they focus on the damping and damping-related force encoding.

Remark 3: The stiffness and damping parameters respectively shown in Fig. 2(a) and (c) are quite large values, i.e., with level 10^5 . It is normal since for the trajectory tracking task with small sampling period, the error terms e and $\dot{e}$ have extremely small values, i.e., with level 10^{-4} , and thus we need large stiffness and damping parameters to result in the suitable control force.

Using the learned VIC with the setting $\epsilon_1 = 1.5$, the assembly experiment results are shown in Fig. 3. Fig. 3(a) shows the velocity tracking result where dashed and solid lines represent the reference and real velocities, respectively. As we can see, the reference velocity can be accurately tracked. The corresponding damping variation is shown in Fig. 3(b). We can find that at the initial stage, the damping will be very small since the initial velocity error $\dot{e}$ is very large. This setting can avoid the generation of the extremely large control force. With the decrease of the $\dot{e}$, the damping continuously increases to ensure a persistent control force. These variation rules are consistent with the desired variable impedance set Ψ shown in Fig. 2(c).

Fig. 3(c) and (d) shows the position tracking result and the corresponding stiffness variations, respectively. Similarly with the case of damping variations, at the first stage, the stiffness for each dimension is negatively related with the position error since the error trajectory in this stage lies on the area away from the origin, see stiffness variation rule in Fig. 2(a). At the last stage, the stiffness decreases as the position error decreases since the error lies in the area near the origin. As the previous discussion, this positively related property between the stiffness and error can suppress the oscillation around the origin.

Fig. 3(e) shows the the distance variations between the robot and goal positions in different dimensions. The final value of these distances, denoted $e_{\text{static},i,f}$ s, determine that whether the assembly task could be accurately executed. The tolerance error for this assembly task is $|e_{\text{static},1,f}| < 0.5 \times 10^{-3}\text{m}$ and $|e_{\text{static},2,f}| < 0.5 \times 10^{-3}\text{m}$. The final error in the z dimension is less important compared with it in the x or y dimension. The final errors in the x and y dimension of this experiment are listed as follows:

$$\begin{cases} |e_{\text{static},1,f}| = 3.12 \times 10^{-5}\text{m} \\ |e_{\text{static},2,f}| = 1.67 \times 10^{-4}\text{m} \end{cases} \quad (61)$$

Thus, the assembly task can be accurately completed.

Fig. 3(f), (g) and (h), respectively, show the control forces in the x , y , and z dimensions. As we can see, the stiffness-related forces are very smooth. Moreover, we can find that the

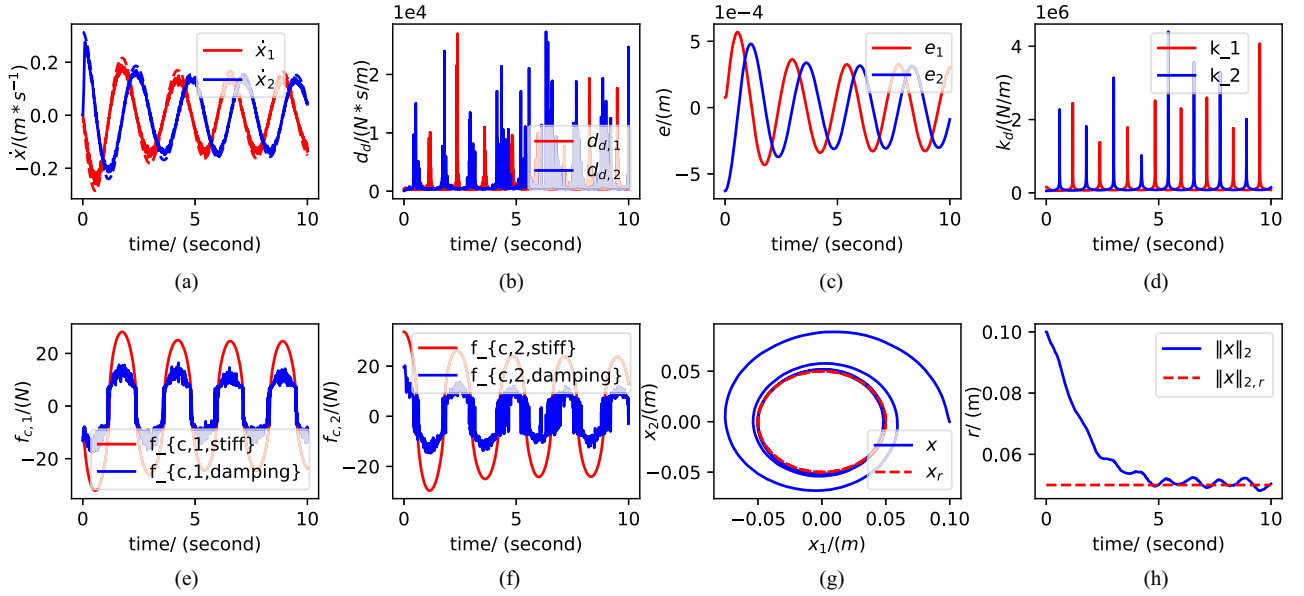


Fig. 5. Results of the robot polish experiment using the proposed VIC with setting $\epsilon_1 = 0.5$. (a) Velocity tracking result. (b) Damping variation. (c) Position tracking error. (d) Stiffness variation. (e) Control force in the x dimension. (f) Control force in the y dimension. (g) 2-dimensional motion trajectory. (h) Distance between the robot and the reference circle center.

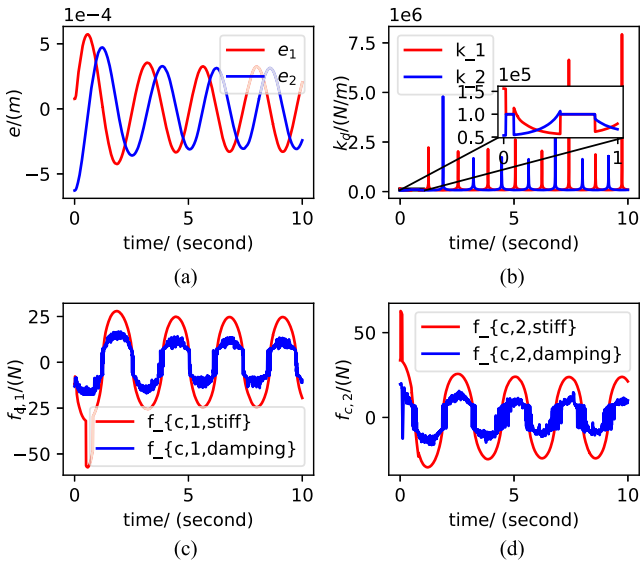


Fig. 6. Results of the robot polish experiment using the tank-based VIC [23]. (a) Position tracking error. (b) Stiffness variation. (c) Control force in the x dimension. (d) Control force in the y dimension.

stiffness-related forces are properly to reduce the position tracking error, which can be reflected by the rapidly increasing and persistent holding of the control force in the initial and middle stages, respectively. This trend is determined by the negatively related property between the stiffness and the tracking error. Comparatively, the damping-related control force is much more noisy due to the noise in the velocity signal and the damping parameters. Actually, before applying the control force on the robot, we will use low-pass filters to smooth these computed signals.

Fig. 3(i) shows the variation of the total energy defined in (12), where V_2 are sums of SNEUMs in each dimension. As we can see, the total energy monotonically decreases to zero, which validates the stability of the robot system.

B. Robot Polish Experiment

In this section, the proposed approach will be validated via the robot polish experiment. Different from the robot assembly task, the robot polish task is a contact-rich task, which might raise more difficulties for accurate tracking. The experiment setting is shown in Fig. 4, where the vector field of the dynamic system planner is shown in the right-upper corner. As a result, the reference velocity will force the robot to conduct periodic motions. The specific design of the planner is omitted for the sake of clarity, and we just denote it as $\dot{x}_r = f(x)$. The reference position is generated as $x_{r,t} = x_t + \dot{x}_{r,t}T$, where $T = 0.002s$ is the sampling period. Similar with the robot assembly experiment, the variable stiffness and damping are also encoded by SNEUMs, and we set $\epsilon_1 = 0.5$ for both stiffness and damping encoding, and other SNEUM parameters are same as these in (60). This setting will generate variable stiffness or damping rules like dashed line in Fig. 2(a), i.e., stiffness and damping values will approach infinity when position and velocity errors approach zero, which could improve the tracking accuracy.

The experiment result is shown in Fig. 5, where Fig. 5(a) and (b), respectively, shows the velocity tracking error and the corresponding damping variations. As we can see, the reference velocity can be well tracked. The damping could be a little bit noisy due to the velocity signal noise.

Fig. 5(c) and (d) shows the position tracking error and the corresponding stiffness variations. The position error trajectories are periodic as what we expect, and the stiffness varies

accordingly. These peaks of the stiffness implies that the position errors approach to zero, and allows the robot to promptly generate control forces in these cases.

Fig. 5(e) and (f), respectively, shows the control forces at the first and second dimensions. As we can see, the control forces are also periodic. The damping-related control force is quite noisy due to the noised velocity signal. Nevertheless, the control performance is quite well, which is reflected in Fig. 5(g), where the robot real motion trajectory (represented by the blue solid line) finally coincides with reference motion trajectory (represented by the red dashed line). Fig. 5(h) gives a more intuitive illustration of this point, where the dashed line represent the radius of reference circle, and the solid line is the real distance of the robot to the circle center. The slight error during the convergence period might be caused by the friction and uncompensated robot dynamics.

We additionally conduct the robot polish experiment using the tank-based VIC approach [22] for comparison. To run a fair comparison, the variable impedance law learned via the proposed approach is set as the original variable impedance law of the tank-based VIC approach. We have carefully tuned the tank parameters to achieve the near-optimal performance. The experiment results are shown in Fig. 6, where Fig. 6(b) shows the variation of the stiffness. Compared with smooth stiffness variation of the proposed approach shown in Fig. 5(d), the stiffness here will sometimes be reshaped to a constant value during the variation process due to the exhaustion of the tank energy. The inappropriate constant stiffness will generate extremely large stiffness-related control forces as shown in Fig. 6(c) and 6(c). As a result, the robot will generate impacts at these moments, which might raise safety issues.

We also conduct comparisons between the proposed VIC approach, constant impedance control (Constant IC), and an online optimization (OP)-based VIC approach [9] with respective to the compliance, accuracy and smoothness metrics. Metrics $|f_{c,i,M}|$ and $|f_{c,i,A}|$, respectively, represent the maximum absolute control force and average absolute control force in i th dimension, and they are used to evaluate the compliance property of the approaches. The metric I_{acc} is used to evaluate the tracking accuracy of the approaches, and it is designed as follows:

$$I_{acc} = T \sum_{t=0}^{t_E} |r - \|x - x_{cen}\|_2| \quad (62)$$

where r is the desired radius of the reference circle, x and x_{cen} are the positions of the robot and reference circle center, respectively. Thus, this metric simultaneously evaluate the dynamic and static tracking errors. The metric $f_{smooth,i}$ is used to measure the smoothness of the approach using the control force variation, which is designed as follows:

$$I_{smooth,i} = \frac{1}{t_E - 1} \sum_{t=1}^{t_E} |f_{c,i,stiffness,t} - f_{c,i,stiffness,t-1}| \quad (63)$$

The comparative results are shown in Table I, where the boldface values represent the best results with respect to the metrics. To conduct a fair comparison, all these approaches are tuned to achieve a similar performance with respect to the I_{acc} to ensure

TABLE I
QUANTITATIVE COMPARISON RESULTS BETWEEN THE PROPOSED VIC (PA), CONSTANT IMPEDANCE CONTROL (IC), AND ONLINE OPTIMIZATION (OP)-BASED VIC [9]

Approaches	Metrics			
	$ f_{c,i,M} $	$ f_{c,i,A} $	I_{acc}	$I_{smooth,i}$
proposed VIC	(49.4, 53.6)	(29.5, 27.3)	0.0841	(0.0861, 0.0845)
Constant IC	(149,266)	(56.1,52.6)	0.0889	(0.221,0.219)
OP-based VIC	(103,205)	(49.5,46.0)	0.0863	(0.424,0.529)

The boldface values indicate the best scores for each metric across these methods.

an accurate tracking. As we can see, the proposed approach achieves the best results with respect to all of these metrics.

IX. CONCLUSION

In this article, we presented a stable state-dependent VIC learning approach based on the energy-function learning framework. The kernel idea is to learn an energy function to serve as a stability certificate and encode the controller's variable stiffness behaviors. We proposed the condition for an energy function to be a proper stability certificate, and designed a corresponding energy function called SNEUM. Based on the SNEUM, we derived two learnable controllers, referred to as the coupled and decoupled variable impedance controllers. By ensuring SNEUM-parameter constraints, the closed-loop robot system will always be GAS and passive. Thorough analyses were conducted for the SNEUM-encoded variable stiffness behaviors, and we showed that the control modes focusing on the control smoothness or tracking accuracy can be mathematically determined by setting a hyper-parameter of the SNEUM. Finally, experiments were conducted to show the effectiveness of the proposed approach.

In this article, due to the stability consideration, the variable stiffness law is only dependent on the position tracking error, which may limit the flexibility. In our future work, we expect to derive the variable stiffness law which can adapt to the contact force.

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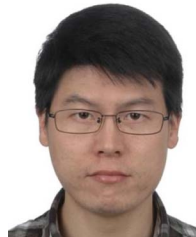
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